



THE ASIAN SCHOOL, DEHRADUN WORKSHEET

Subject : SCIENCE (BIOLOGY)

Chapter:- Cell: The building block of life

Topic: Cell structure

Class : IX
DDH

MM : 20

Teacher :

	SECTION-A	
1.	The cell membrane is mainly responsible for: A. Producing energy B. Controlling the movement of substances into and out of the cell C. Making proteins D. Storing genetic material	1
2.	The jelly-like substance present between the cell membrane and nucleus is called: A. Cell wall B. Cytoplasm C. Nucleoplasm D. Vacuole	1
3.	Which structure contains the genetic material of a cell? A. Cytoplasm B. Cell membrane C. Nucleus D. Cell wall	1
4.	The cell membrane is described as selectively permeable because it: A. Allows all substances to pass through it B. Does not allow any substance to pass through it C. Allows only certain substances to pass through it D. Allows only water to pass through it	1
5.	When a cell is placed in a solution having a higher concentration of solute than the cell sap, water moves out of the cell. This process is called: A. Diffusion	1

	B. Osmosis C. Plasmolysis D. Endocytosis	
7.	Differentiate between diffusion and osmosis on the basis of the movement of particles and the requirement of a selectively permeable membrane. Give one suitable example of each process.	2
	SECTION-C	
8.	A plant cell is placed in a concentrated salt solution. A. What will happen to the movement of water? B. What will happen to the cell membrane and cytoplasm. C. Name the process responsible for this change.	3
9.	Explain how the nucleus controls the activities of a cell. Also mention the role of chromosomes present inside the nucleus.	3
	SECTION-D	
10	The cell membrane, cytoplasm and nucleus work together to maintain the proper functioning of a cell. Explain this statement by describing: A. The structure and function of the cell membrane. B. The role of cytoplasm in cellular activities. C. The structure and functions of the nucleus. D. How the nucleus and cytoplasm together help in controlling and carrying out cellular activities.	5



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SECTION-A

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1.	Which cell organelle is known as the “powerhouse of the cell”? A. Ribosome B. Mitochondria C. Golgi apparatus D. Vacuole	1
2.	Which organelle controls most of the activities of the cell? A.Lysosome B. Vacuole C. Cytoplasm D. Nucleus	1
3.	Which organelle is responsible for protein synthesis? A. Mitochondria B. Ribosome C. Golgi apparatus D. Chloroplast	1
4.	Which organelle is mainly involved in packaging and secretion of substances? A. Golgi apparatus B. Nucleus C. Vacuole D. Lysosome	1
5.	A cell contains many chloroplasts and a large central vacuole. Which type of cell is it most likely to be? A. Animal cell	1

	B. Plant cell C. Bacterial cell D. Red blood cell	
	SECTION-B	
6.	What is the function of lysosomes? Why are they sometimes called the “suicidal bags” of the cell?	2
7.	Differentiate between rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) on the basis of their structure and function.	2
	SECTION-C	
8.	A cell is unable to produce enough energy even though It has sufficient food and oxygen. Which cell organelle may not be functioning properly? Identify the organelle and explain how its structure is related to its function.	3
9.	Explain how the endoplasmic reticulum, Golgi apparatus and lysosomes work together to process, transport and manage materials inside a cell.	3
	SECTION-D	
10	A student examines two different types of cells under a microscope. Cell A possesses a cell wall, chloroplasts and a prominent central vacuole, whereas Cell B lacks these structures. Answer the following: A. Identify Cell A and Cell B on the basis of their structural features. B. Name the cell organelle that carries out photosynthesis in Cell A. C. Mention any two functions performed by the large central vacuole in Cell A. D. Describe the role of the nucleus in regulating the activities of both cells. E. Explain how the presence of a cell wall contributes to the structural stability and protection of Cell A.	5



WORKSHEET

Chapter:- Cell: The building block of life

Topic: Cell Division

Class : IX
DDH

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SECTION-A

	SECTION-A	
1.	The process by which a parent cell divides to form new cells is called: A. Respiration B. Cell division C. Nutrition D. Excretion	1
2.	Which biomolecule is the main source of energy for our body? A. Proteins B. Carbohydrates C. Vitamins D. Minerals	1
3.	Proteins are mainly required by the body for: A Growth and repair B. Providing only water C. Maintaining body temperature D. Producing oxygen	1
4.	The cell membrane is described as selectively permeable because it: A. Allows all substances to pass through it B. Does not allow any substance to pass through it C. Allows only certain substances to pass through it D. Allows only water to pass through it	1
5.	During cell division, the genetic material must be properly distributed to the daughter cells so that: A. They can lose their chromosome. B. They receive the required genetic information C. They become non-living D. They stop performing all functions	1
	SECTION-B	
6.	What is cell division? State two reasons why cell division is important for living organisms.	2

7.	Identify any two biomolecules found in cells and mention one function of each.	2
	SECTION-C	
8.	A cell undergoes division and produces two daughter cells. Explain why DNA replication and proper distribution of chromosomes are important before and during cell division.	3
9.	A group of cells in a tissue starts dividing continuously without normal control. A. What condition may develop due to such uncontrolled cell division? B. What is meant by cancer? C. Why can uncontrolled cell division be harmful to the body?	3
	SECTION-D	
10	Cells require both cell division and biomolecules for their growth and functioning. Explain the following: A. Why is cell division necessary for growth and repair B. What is the role of carbohydrates in cells? C. What is the role of proteins in the body? D. What is meant by uncontrolled cell division? E. How can uncontrolled cell division lead to cancer?.	5



THE ASIAN SCHOOL, DEHRADUN WORKSHEET

Chapter:- Tissues in Action
Topic: Plant Tissues

Class : IX
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MM : 20

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	SECTION-A	
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1.	Which of the following tissues consists of living cells and generally stores food? A. Sclerenchyma B. Parenchyma C. Xylem D. Epidermis	1
2.	The tissue that provides flexibility to young stems and leaf stalks is: A. Parenchyma B. Collenchyma C. Sclerenchyma D. Internode	1
3.	The epidermis of a plant primarily helps in: A. Transporting food B. Protecting the plant body C. Producing seeds D. Conduction of water	1
4.	A plant part contains cells with unevenly thickened walls, particularly at the corners. Which tissue is most likely present? A. Parenchyma B. Collenchyma C. Sclerenchyma D. Meristematic tissue	1
5.	A student observes a tissue made up of dead cells with thick, lignified walls and very little intercellular space. Identify the tissue and state its major function. A. Parenchyma – storage B. Collenchyma – flexibility C. Sclerenchyma – mechanical strength D. Epidermis – gaseous exchange	1
	SECTION-B	
6.	Give two reasons why a plant tissue is different from animal tissue.	2
7.	What is ground tissue? Mention any one function of ground tissue.	2
	SECTION-C	

8.	Differentiate between Xylem and Phloem on the basis of: A. Any two components B. Functions C. Direction of conduction	3
9.	What is supportive tissue? Explain how supportive tissues are modified to provide mechanical strength and support to plants. Give two examples.	3
	SECTION-D	
10	A plant grows continuously throughout its life because certain regions contain cells that retain the ability to divide repeatedly. These tissues produce new cells, which later differentiate and contribute to the growth and development of various plant parts. Answer the following: A. Name the tissue responsible for the growth of plants. B. Describe the structure of meristematic cells, including any two features that distinguish them from mature permanent cells. C. Name the type of meristematic tissue located at the growing tips of roots and shoots, and state the kind of growth it produces. D. Explain one function of meristematic tissue in the development of a plant.	5



THE ASIAN SCHOOL, DEHRADUN WORKSHEET

Chapter:- Tissues in Action
Topic: Animal Tissues

Class : IX

Teacher : DDH

MM : 20

	SECTION-A	
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1.	Which tissue forms the protective covering of most organs and body surfaces? A. Connective tissue B. Muscular tissue C. Epithelial tissue D. Nervous tissue	1
2.	Which connective tissue connects muscle to bone? A. Ligament B. Tendon C. Cartilage D. Areolar tissue	1
3.	Which type of muscle is involuntary and found in the walls of internal organs such as the intestine? A. Striated muscle B. Smooth muscle C. Cardiac muscle D. Skeletal muscle	1
4.	A tissue is made up of elongated cells that contract and relax to produce movement. Which tissue is being described? A. Nervous tissue B. Epithelial tissue C. Muscular tissue D. Connective tissue	1
5.	A person accidentally touches a hot object and immediately withdraws the hand. Which property of nervous tissue enables this rapid response? A. Secretion B. Contraction C. Transmission of electrical impulses D. Storage of fat	1
	SECTION-B	
6.	Explain fluid connection tissue. Mention one important function of epithelial tissue.	2
7.	Compare between a tendon and a ligament on the basis of their structures and function.	2
	SECTION-C	

8.	Epithelial tissues have different structures to perform different functions. Compare squamous epithelium and columnar epithelium on the basis of their structure, location and function. How does their structure help them perform their functions.	3
9.	Differentiate between squamous epithelium and columnar epithelium on the basis of their structure, location and function.	3
	SECTION-D	
10	The human skeleton is not merely a framework for the body; different parts are specially adapted to provide protection, support and movement. Analyse the role of the following structures: A. The rib cage is made up of ribs and the sternum. Explain how its structure enables it to protect vital organs while allowing the movements necessary for breathing. B. The skull protects the brain, but its bones are not all freely movable. Explain how the structure and arrangement of the skull provide effective protection to the brain. C. Why is the vertebral column made up of several vertebrae instead of being a single rigid bone? D. What is the basic structural and functional unit of nervous tissue?	5